Kinematics

Displacement, velocity, and acceleration

1 A particle moves with displacement $s = t^3 - 6t^2 + 9t$ (in meters, t in seconds). Find expressions for its velocity v and acceleration a.

2 A particle moves with displacement $s = 2t^3 - 9t^2 + 12t$ (in meters, t in seconds). Find its velocity at $t = 2$ seconds.

The suvat equations for constant acceleration

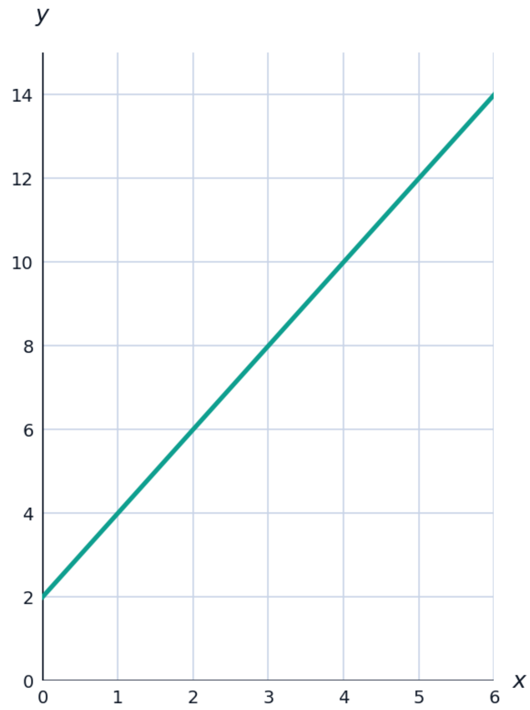
3 A car starts from rest and accelerates uniformly at 2.5 m/s^2 for 8 seconds. Find its final velocity and the distance traveled, using $v = u + at$ and $s = ut + \frac{1}{2}at^2$.

4 A ball is thrown upward with initial velocity $u = 15$ m/s. Using $g = 9.8$ m/s^2 (deceleration), find the maximum height reached, using $v^2 = u^2 + 2as$ with $v = 0$ at the top.

5 A cyclist decelerates uniformly from 12 m/s to rest over a distance of 36 meters. Find the deceleration, using $v^2 = u^2 + 2as$.

Velocity-time graphs

6 A particle accelerates uniformly from $u = 2$ m/s to $v = 14$ m/s over 6 seconds, then travels at constant velocity for a further 4 seconds. The velocity-time graph is shown. Find the total distance traveled, using the area under the graph.



7 A particle decelerates uniformly from $v = 20$ m/s to rest over 5 seconds, giving a velocity-time graph that is a straight line from $(0, 20)$ to $(5, 0)$. Find the distance traveled, using the area under the graph.

Motion under gravity

8 A stone is dropped from a height of 45 meters. Using $g = 9.8$ m/s^2, find the time taken to reach the ground, and its speed on impact.

9 A ball is thrown vertically upward with initial speed $u = 24.5$ m/s. Using $g = 9.8$ m/s^2, find the total time taken to return to its starting point.

Using calculus for variable acceleration

10 A particle has velocity $v = 3t^2 - 2t$ (m/s). Find its displacement from $t = 0$ to $t = 3$, and its acceleration at $t = 3$.

11 A particle has velocity $v = 4t - t^2$ (m/s). Find its displacement from $t = 0$ to $t = 2$, and its acceleration at $t = 2$.

Finding when a particle is at rest or changes direction

12 A particle has velocity $v = t^2 - 5t + 6$ (m/s). Find the times at which the particle is instantaneously at rest.

13 A particle has velocity $v = t^2 - 7t + 10$ (m/s). Find the times at which the particle is instantaneously at rest.

Relative and combined motion

14 Two cars start 300 meters apart on a straight road and travel toward each other, one at 18 m/s and the other at 12 m/s, both at constant speed. Find the time until they meet.

15 Two cars start at the same point on a straight road, one traveling at 25 m/s. The second car starts 4 seconds later, traveling at 30 m/s. Find how long after the first car starts the second car catches up.

A well-designed word problem: a train between two stations

16 This question has two parts. A train starts from rest at station A and accelerates uniformly at 0.5 m/s^2 until it reaches a speed of 20 m/s. It then travels at this constant speed before decelerating uniformly at 1 m/s^2 to come to rest at station B. The total distance between the stations is 2400 meters. (a) Find the time and distance covered during the acceleration phase. (b) Find the total journey time between the two stations.